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Code JupyterLab Python 3 (ipykernel)

[127]:
```
import numpy as np
a=np.array([[1,2],[22,44]])
b=np.array([[4,5],[56,65]])
print(np.dot(a,b))
print(a@b)
print(a.T)
print(b.T)
```

```
[[ 116  135]
 [2552 2970]]
[[ 116  135]
 [2552 2970]]
[[ 1 22]
 [ 2 44]]
[[ 4 56]
 [ 5 65]]
```

[132]:
```
np.random.seed(30)
print(np.random.rand(5,5))
print(np.random.randn(3,2))
print(np.random.randint(1,10,size=(3,3)))
```

```
[[0.64414354 0.38074849 0.66304791 0.16365073 0.96260781]
 [0.34666184 0.99175099 0.2350579  0.58569427 0.4066901 ]
 [0.13623432 0.54413629 0.51817635 0.76685511 0.93385014]
 [0.08970338 0.19577126 0.99419368 0.2351805  0.23898637]
 [0.62909983 0.73495258 0.68834438 0.03113075 0.90251384]]
[[ 0.45579259 -1.7503454 ]
 [-0.28541593 -0.07451087]
 [ 1.26657506 -0.02213995]]
[[8 5 7]
 [9 7 1]
 [8 6 4]]
```

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Code

JupyterLab Python 3 (ipykernel)

[138]:
```
print(np.vstack((a,b)))
print(np.hstack((a,b)))
print(np.concatenate((a,b)))
print(np.split(np.arange(20),5))
```

```
[[ 1  2]
 [22 44]
 [ 4  5]
 [56 65]]
[[ 1  2  4  5]
 [22 44 56 65]]
[[ 1  2]
 [22 44]
 [ 4  5]
 [56 65]]
[array([0, 1, 2, 3]), array([4, 5, 6, 7]), array([ 8,  9, 10, 11]), array([12, 13, 14, 15]), array([16, 17, 18, 19])]
```

[141]:
```
a = np.array([2,3,461,455,63,6,0,2])
print(np.sort(a))
print(np.argsort(a))
print(np.where(a>3))
```

```
[  0   2   2   3   6  63 455 461]
[6 0 7 1 5 4 3 2]
(array([2, 3, 4, 5]),)
```

[146]:
```
print(a)
b = a>3
print(b)
print(a[b])
```

```
[  2   3 461 455  63   6   0   2]
[False False  True  True  True  True False False]
[461 455  63   6]
```

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[148]:

```
np.save('arrary_data.npy',a)
data = np.load('array_data.npy')
print(data)
```

```
[1 2 1]
```